

Physics

1. (4) + 0.5 ✓

2. (2) outer surface of cornea. ✓

3. (4) virtual, erect & enlarged. ✓

Explanation

4. (A) Both A & R are true & R is the correct reason of A.

5. (B) Assertion is true but reason is false. ✓

A-6. 2 Common defects of vision :-

(i) Myopia → In this defect a person cannot see distant object clearly, also known as Near Sightedness.

Correction: He/she can wear concave lens.

(ii) Hypermetropia: In this defect one can't see clearly the objects closer to eye, also known as Far sightedness.

Correction: One should wear convex lens.

A-7 (a) A Pencil when immersed in water appears bent & short because of the refraction of light. If an object is in denser medium observer is in rarer medium then the apparent height is less than the real height.

Q. 7. (6) Given

object distance = $u = ?$

magnification = $m = 20$

$v = -20 \text{ cm}$

$f = ?$

now using magnification formula

$$m = \frac{v}{u}$$

$$\Rightarrow 20 = \frac{-20}{u}$$

$$\Rightarrow \underline{u = -10 \text{ cm}}$$

8. Given,

object distance $u = -10 \text{ cm}$, focal length = 15 cm ,

image distance = $v = ?$

now using mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\Rightarrow \frac{1}{15} = \frac{1}{v} + \frac{1}{(-10)}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{15} - \frac{1}{10} = \frac{2-3}{30}$$

$$\Rightarrow \underline{v = -6 \text{ cm}}$$

The image distance is negative that means image is Virtual & Erect & as object is in between O & F hence it is enlarged image. as it is convex mirror the image is diminished.

$$-\frac{1}{v} = -\frac{1}{15} - \frac{1}{10} \quad f v = +30 \text{ cm}$$

$$-\frac{2-3}{30} = -\frac{1}{v} - \frac{1}{30}$$

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9. Given.

$u = -10 \text{ cm}$, forms real image, $m = -3$

$v = ?$, $f = ?$

as it's forming real image, hence it's a convex lens, & $m = -3$

now, using magnification formula.

$$m = \frac{v}{u}$$

$$\Rightarrow -3 = \frac{-10}{v}$$

$$-3 = \frac{v}{-10}$$

$$\Rightarrow v = \frac{-10}{-3}$$

$$\Rightarrow v = 30 \text{ cm}$$

$$\Rightarrow v = \frac{10}{3} \text{ cm}$$

image is formed at 30 cm behind the lens.

now using lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{30} + \frac{1}{10} = \frac{1+3}{30} = \frac{4}{30}$$

$$f = \frac{30}{4} = 7.5 \text{ cm}$$

10 far point = $x = 80 \text{ cm}$, $f = ?$,

⇒ formula for focal length of lens for myopic lens

⇒ $f = -x \text{ cm}$,
 $f = -80 \text{ cm or } -0.8 \text{ m}$

Power of lens

⇒ $P = \frac{1}{f} =$

⇒ $P = \frac{1}{-0.8}$

⇒ $P = -1.25 \text{ D}$

∴ the lens is concave lens.

11 (a) Conver lens

(b) $\frac{v}{u}$ or magnification will be negative because it's a real & inverted image.

(c) Given, $f = 20 \text{ cm}$, $u = -21 \text{ cm}$, $v = ?$

using lens formula

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

⇒ $\frac{1}{20} = \frac{1}{v} + \frac{1}{21}$

⇒ $\frac{1}{20} - \frac{1}{21} = \frac{1}{v}$

20
2x2x5

21
3x4

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$$\Rightarrow \frac{1}{v} = \frac{81-20}{420}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{420}$$

$$\Rightarrow v = 420 \text{ cm or } 4.2 \text{ m}$$

the screen must be placed 4.2m or 420cm away from the lens.

12. (a) Laws of Refraction of Light

(i) The incident ray, the normal to the refracting surface at the point of incidence and the refracted ray, all lie in the same plane.

(ii) The ratio of sin of angle of incidence is equal to sin of angle of refraction for 2 given media
 $\Rightarrow \sin i = \sin r$

Absolute Refractive Index = $\frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$

$$\Rightarrow n = \frac{c}{v}$$

(b) ~~refraction in~~

Phy

12 (6) Given with respect to
 Refractive index of water to Air = $\frac{4}{3}$

Refractive index of glass to Air = $\frac{3}{2}$

Speed of light in glass = $2 \times 10^8 \text{ m s}^{-1}$

to find (i) speed of light in water

Let the speed of light in water be S_1 , in air S_2 &
 glass S_3

$\Rightarrow$ Refractive index of glass to air = $\frac{3}{2} = \frac{S_2}{S_3}$

$\Rightarrow \frac{3}{2} = \frac{S_2}{2 \times 10^8 \text{ m s}^{-1}}$

$\Rightarrow \frac{3 \times 2 \times 10^8 \text{ m s}^{-1}}{2} = S_2$

$\Rightarrow$ speed of light in air = $S_2 = 3 \times 10^8 \text{ m s}^{-1}$

now, Refractive index of water to air = $\frac{4}{3} = \frac{S_2}{S_1}$

$\Rightarrow \frac{4}{3} = \frac{3 \times 10^8 \text{ m s}^{-1}}{S_1}$

$\Rightarrow 4S_1 = 9 \times 10^8 \text{ m s}^{-1}$

$\Rightarrow S_1 = \frac{9}{4} \times 10^8 \text{ m s}^{-1} \text{ or } 2.25 \times 10^8 \text{ m s}^{-1}$

now, speed of light in water = $2.25 \times 10^8 \text{ m s}^{-1}$ or $\frac{9}{4} \times 10^8 \text{ m s}^{-1}$